

Lab 4: Direct/Global Separation

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1 Separation

		
Floodlit	Direct	Global

Scene	A tabletop corner on a nightstand with a wood-grain top and silver metal edging, in front of a textured white wall and a white door frame. The back wall is a sheet of white paper, and a sheet of orange paper is folded at 90° to it. Objects: a red apple (glossy, waxy skin); a candle in a clear glass jar with a paper label and a polished metal lid; a white glazed ceramic mug; a plush rooster toy made of felt and fabric; and a yellow padded paper mailing envelope lying flat in the foreground.
Capture	<u>Light source, how it was swept, camera settings, number of frames.</u>

2 Features *written by a group member, no AI*

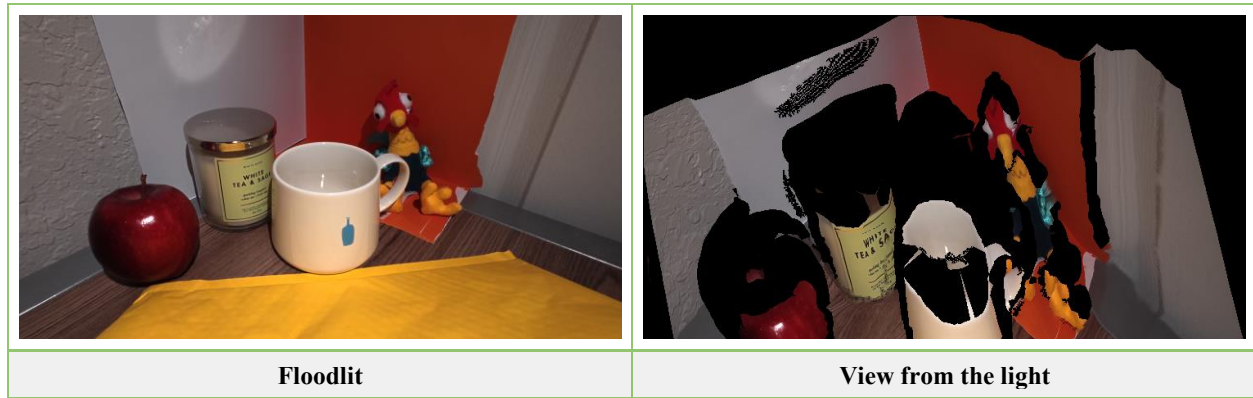
- **Specularities:** The specularities in the direct image appear on the front upper portion of the apple, where it is the shiniest and reflecting light and appears because the apple has a waxy/glossy surface. Another specularity is on the side of the lid of the candle, which is highlighted because of the metallic surface. One other specularity is the bright spots on the inside and handle of the mug, which happens because the mug material is reflective.
- **Shadows:** The direct image is dark underneath and behind both the mug and the candle. It is also dark in the space where the mug overlaps the candle, due to the shadow being cast by the mug. The apple has the darkest edges/shadows, with them being behind, underneath, and on the left edge. The left and right corners of the image also have shadowed edges. Lastly, the chicken casts a shadow on the back left of it. All these shadows occurred because of the direct light hitting one side and the objects obscuring that light, effectively causing the shadows behind and around them.
- **Interreflections:** The light that bounced off the orange wall and yellow-orange package give the global image an orange-reddish tint. This is especially noticeable on the mug, which tones look much warmer than before. The light that bounced off the white paper makes the candle appear lighter in the photo, and the light that bounced off the orange/white walls helped fill in the shadows. This is very noticeable behind the chicken which essentially no longer has a shadow and is just filled in with orange.

3 AI separation



Tool and prompt	Tool: We used ChatGPT's image generation. We uploaded only our floodlit image and asked for the direct and global images in two separate requests, both starting from the same floodlit photo. Direct prompt: "This is a photo of a scene lit by a single light source. Generate the DIRECT illumination component of this image: only light that travels from the light source to a surface and straight to the camera (one bounce). Keep the exact same viewpoint, framing and objects. Remove all indirect light (interreflections between surfaces, subsurface scattering). Keep specular highlights. Shadowed regions should be dark." Global prompt: "This is a photo of a scene lit by a single light source. Generate the GLOBAL illumination component of this image: only light that reaches each point after bouncing off other surfaces or scattering inside materials (interreflections, subsurface scattering). Keep the exact same viewpoint, framing and objects. Remove all direct illumination and specular highlights. Keep the same exposure as the input; do not brighten the result."
What confused it	The AI treated the photo as a normal room shot and relit it instead of separating it. Its direct image adds hard black shadows that do not exist in the scene (the candle's shadow on the white paper, the apple's shadow on the wall, a dark crescent inside the mug) and darkens the whole right side, as if a strong light came from the left. It also removed the flashlight's bright spot on the white paper, which is the direct light pattern itself. Its global image is far too bright (38% of floodlit in linear light vs. our measured 6%), because it assumed ambient room light: the textured wall, door frame and apple stay clearly visible, while in our measurement they are nearly black because the only light source is a small flashlight in a dark room. Its direct + global adds up to 117% of the input, not 100%. It did get the yellow bounce from the envelope onto the lower mug right. What confused it: a single photo cannot show how much of the light is indirect, and our scene (one small handheld light, dark surroundings, a visible hotspot) does not match the evenly lit rooms the AI expects.

4 Extra credit: light's view



We followed the perspective shadow camera from Koppal & Narasimhan (Sec. 2.1 and 3). We kept the flashlight and camera fixed and swept an opaque occluder in front of the light twice: once moving left to right (sweep_h, 795 frames) and once moving top to bottom (sweep_v, 789 frames). For every camera pixel, we found the time when the shadow passed over it. We converted the video to linear brightness, divided out the flicker of the hand-held light, smoothed each pixel's brightness over 5 frames, and took the weighted centre of its darkest stretch. This gives a time t_h from the first sweep and t_v from the second. Pixels whose brightness never dropped by at least 60%, or that sat on object edges, were thrown out as unreliable. The two times identify one ray leaving the light, so (t_h, t_v) is the pixel's position in a virtual camera at the light. By Helmholtz reciprocity, $I_{\text{light}}(t_h, t_v) = I_{\text{camera}}(x, y)$, so we moved each floodlit pixel to that position. We used a Delaunay triangulation to fill in between pixels, and rotated the result by 12.7° because both sweeps were slightly tilted. The result is a 737×577 image.

As in the paper's Fig. 3, the view now looks down from the upper left, and the cast shadows in our floodlit image are gone, because the light cannot see its own shadows. The large black regions next to each object are the reverse case: parts of the scene the light can see but our camera cannot. They are exactly where the objects would cast shadows if the light were at the camera. We left them black instead of filling them in. Unlike the paper, we swept the occluder by hand and did not use a calibration plane, so we assumed a constant sweep speed (one frame = one light-view pixel). Uneven hand speed causes some stretching, and the moving hand-held light makes object edges ragged.

5 AI disclosure

- Chengzhen Jiang: AI tools this member used and for what.
- Yanru Li: Used ChatGPT's image generation to create the AI direct and AI global images in Section 3.
- Ariana Delamata: